AI-Driven Health Analyzer Real-Time Physical and Mental Wellness Optimization using ML

Project ID 25-26J-225

Project Proposal Report

Kavilakshan Vijayakumar
IT22116338

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

Department of Information Technology

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DECLARATION

I affirm that this is my original work and that, to the best of my knowledge and belief, it does not contain any previously published or written material by anyone else, with the exception of instances in which credit is provided within the text. Nor does it incorporate, without acknowledgement, any previously submitted material for a degree or diploma at any other university or institute of higher learning.

Group Member Name	Student ID	Signature
Kavilakshan V	IT22116338	V. Kavilakshan.

Signature of the Supervisor
(Dr. Kapila Dissanayaka)

Date

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ABSTRACT

Mental Health Assessmemt focuses on the development of an AI-based system for detecting and managing emotional states through video analysis and personalized interventions. The core of the system involves analyzing user-uploaded videos to predict emotional states such as sadness, happiness, anger, or stress. Using advanced deep learning models, the system processes both facial expressions and voice tone to provide an accurate emotional assessment. Upon detecting the emotional state, the system offers personalized suggestions to help users manage their emotions, including video links, coping strategies, or other helpful resources tailored to the identified emotional state.

Additionally, a mobile application will complement the system by predicting the user's emotional state through daily facial video uploads. This app will track the user's emotional progress over time and assist in recovery through features like reminders, personalized questions, and continuous emotional support. The app will also encourage users to upload new videos regularly to ensure accurate tracking and provide ongoing emotional regulation support. By combining real-time emotion detection, personalized intervention strategies, and mobile app integration, this project aims to create a comprehensive tool for emotional well-being that leverages AI to promote mental health management intervention.

Keywords:

Artificial Intelligence (AI), Emotional State Detection, Video Analysis, Facial Expression Recognition, Voice Tone Analysis, Deep Learning Models, Emotion Prediction, Personalized Interventions, Emotional Regulation, Mobile Application, Daily Video Uploads, Emotional Progress Tracking, Coping Strategies, Real-time Emotion Detection, Mental Health Management, AI-based Mental Health Support, Personalized Emotional Support, User Engagement, Emotional Recovery.

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1.INRODUCTION

The increasing integration of Artificial Intelligence (AI) into everyday life has led to significant advancements in various fields, including health and well-being. This project focuses on three key components: **hydration management**, **workout optimization**, and **mental health assessment**. The **hydration management** component aims to provide personalized recommendations for fluid intake, helping users maintain optimal hydration levels. The **workout optimization** component focuses on designing personalized exercise plans that adapt to the user's fitness level, goals, and progress, ensuring effective and safe workouts. The **mental health assessment** component, which is the focus of this project, utilizes AI technologies to detect and regulate emotional states, promoting mental well-being.

The **mental health assessment** component of the project develops an AI-driven system that utilizes facial expression recognition and voice tone analysis to detect a user's emotional state in real-time. By analyzing these visual and auditory cues, the system can predict emotions such as happiness, sadness, anger, or stress, providing a more accurate and personalized approach to emotion recognition. Once the emotional state is identified, the system offers tailored suggestions for emotional regulation, which may include coping strategies, video links, or other resources that address the individual's specific emotional needs. This system is complemented by a mobile application that tracks the user's emotional progress through daily video uploads, offering real-time emotional support and continuous monitoring. Features such as reminders, personalized questions, and feedback mechanisms encourage users to engage with their emotional health, creating a seamless and supportive experience for managing mental well-being. By combining real-time emotion detection, personalized interventions, and mobile app integration, this project aims to create an innovative and accessible tool for emotional health management.

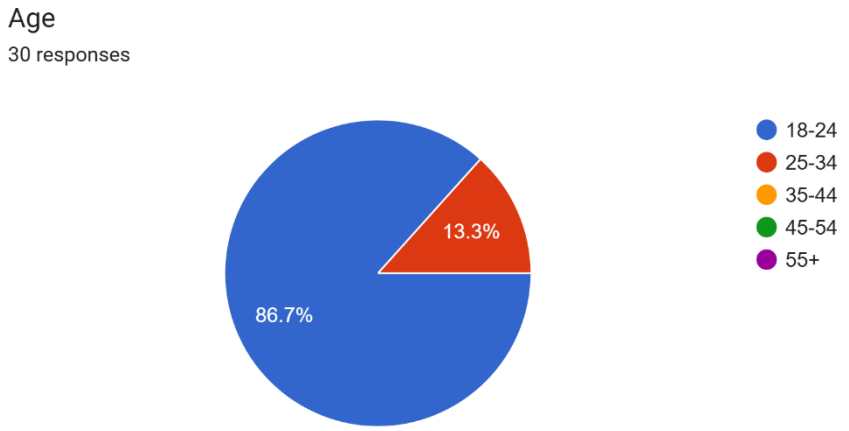

Figure 1: survey 1

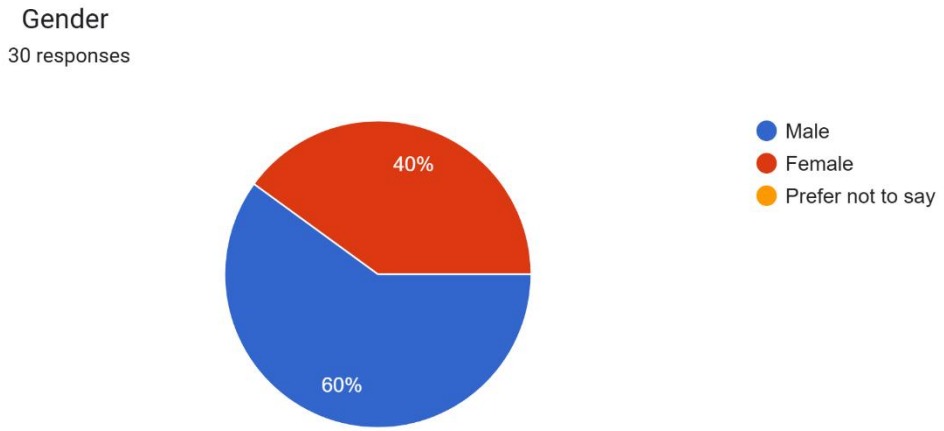

Figure 2 : survey 2

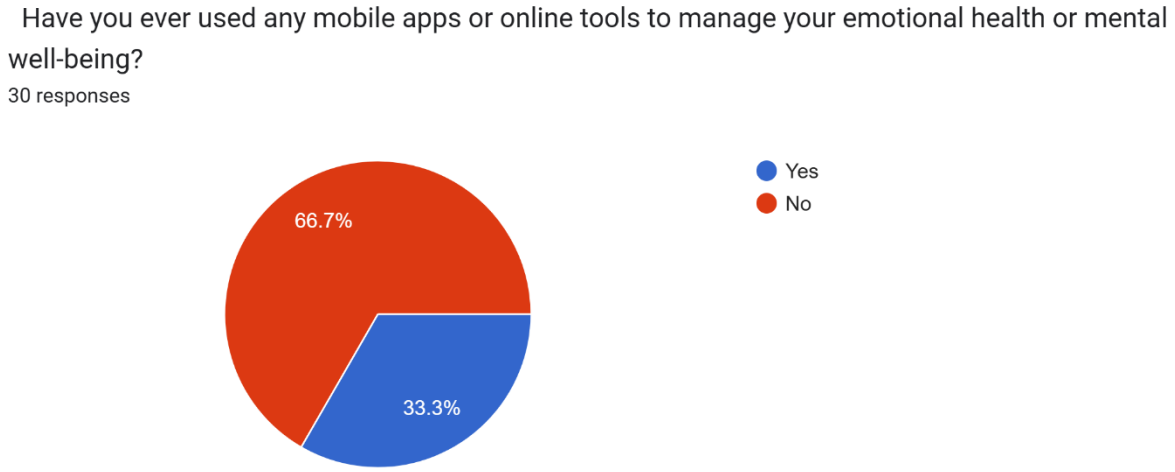

Figure 3 : already used app

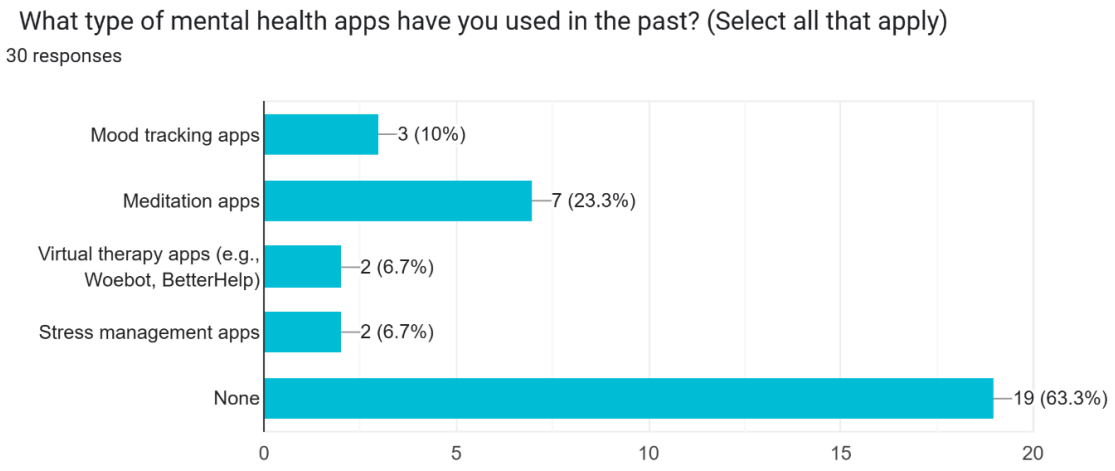

Figure 4 : Type of mental health

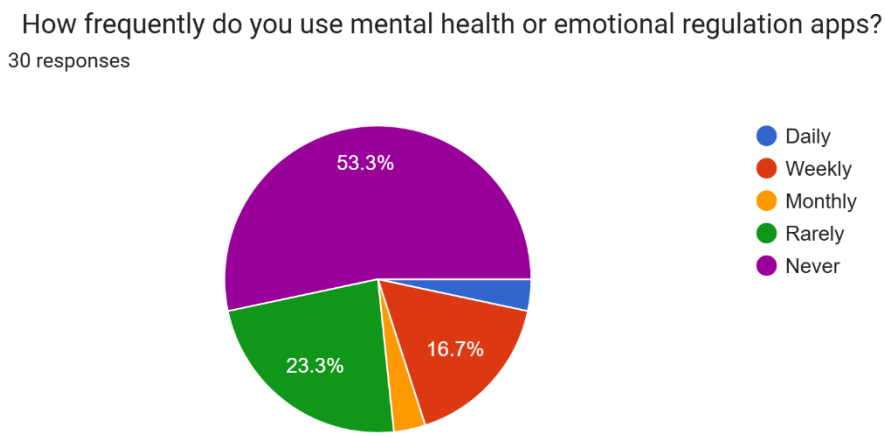

Figure 5 : frequently used

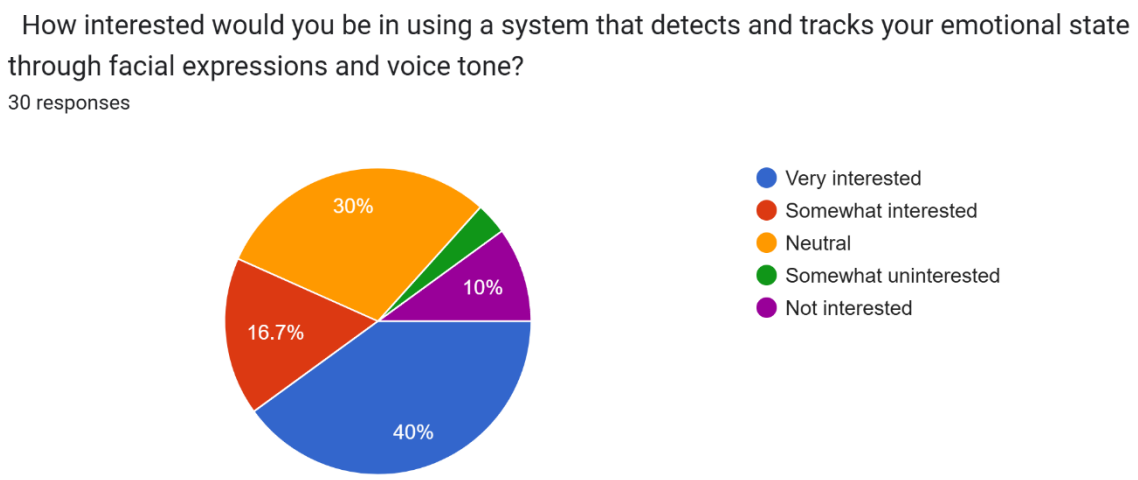

Figure 6 : how intereted

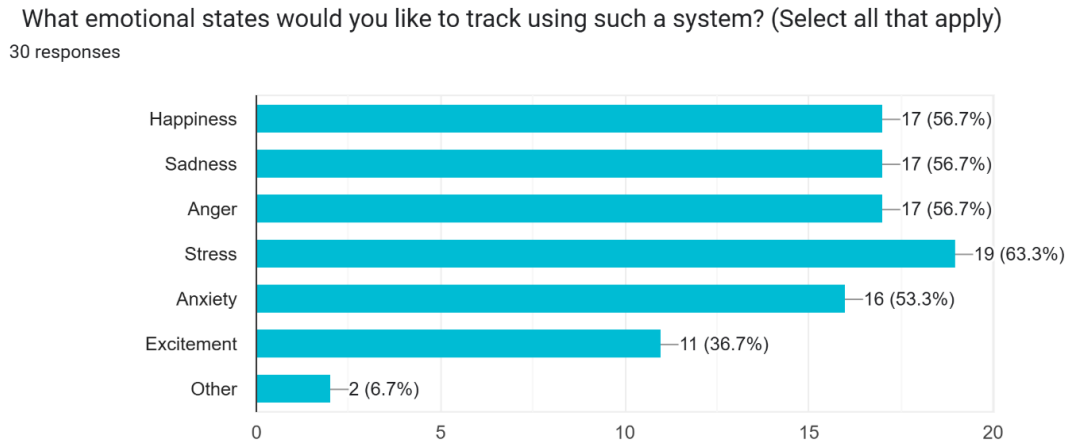

Figure 7: emotional states

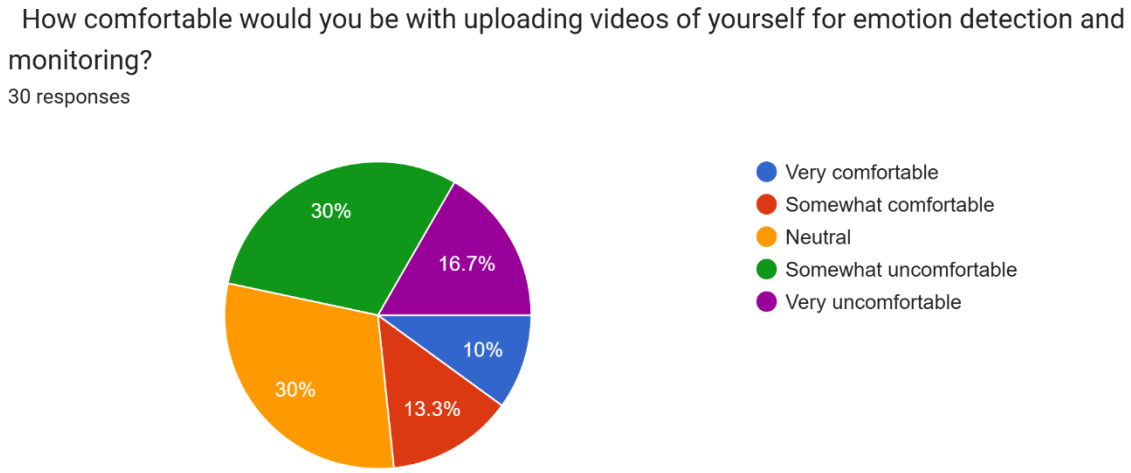

Figure 8: comfortable

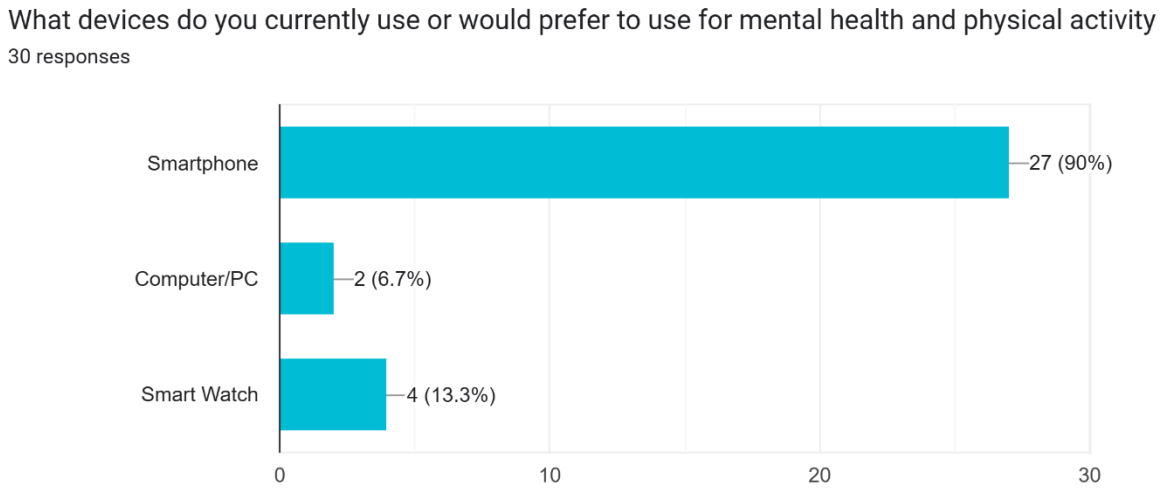

Figure 9: devices

1.1 Background & Literature Survey

1.1.1Background

In recent years, the integration of Artificial Intelligence (AI) in mental health applications has gained significant attention, especially in the context of emotional well-being. Traditional methods of emotional state detection, such as self-reporting or therapist evaluations, can be subjective and limited in their ability to monitor real-time emotional changes. With the advent of AI technologies, particularly deep learning models, it has become possible to assess and manage emotions through more objective and continuous approaches. AI can analyze both visual and auditory cues from videos and voice recordings to provide a more accurate understanding of a person's emotional state.

Facial expression recognition and voice tone analysis are among the most effective techniques used for emotion detection. These methods have the potential to revolutionize emotional health management by offering real-time, personalized interventions. By combining these modalities, the project aims to enhance the precision of emotion detection, ensuring that emotional states are recognized more accurately, regardless of the environment or context.

Additionally, the growing interest in mobile applications for mental health has led to the development of tools that monitor emotional well-being and provide real-time feedback. Mobile apps that track emotional progress through video uploads, reminders, and coping strategies are becoming increasingly popular, offering users a convenient way to engage with and improve their mental health.

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1.1.2 Literature Survey

Facial expression recognition and voice tone analysis are critical for emotion detection in AI systems. Research by Ekman and Friesen (1978) established six universal emotions visible on the face, which are detected using deep learning models like Convolutional Neural Networks (CNNs), as demonstrated by Zhu et al. (2017) and Scherer (2019). Voice tone, including pitch and tempo, also plays a key role in detecting emotions, with advancements in Long Short-Term Memory networks (LSTMs) improving recognition accuracy. AI systems, such as Woebot (Fitzpatrick et al., 2017), offer real-time emotional regulation by providing personalized suggestions based on detected emotions. Mobile apps like Happify and Moodpath track mood and offer activities for emotional well-being. However, there are challenges in integrating real-time detection, cultural sensitivity, and addressing privacy concerns, as highlighted by Binns (2018) and Poria et al. (2017). These systems need to account for cultural variations and ensure ethical handling of emotional data.

1.2 Research Gap

Despite advancements in AI-driven emotion detection, several gaps exist. One key gap is the **integration of multimodal data**, as current systems typically analyze facial expressions or voice tone separately. Combining both could enhance detection accuracy, but challenges remain in processing them together in real time. Additionally, while AI systems like Woebot offer emotional support, there is a need for **real-time, personalized interventions** that adapt to changing emotional states. **Cultural sensitivity** is another gap, as many models fail to account for variations in emotional expression across cultures, leading to biases. Ethical concerns, especially around **privacy and data security**, are critical, particularly when handling sensitive emotional data. Finally, there is limited research on **long-term emotional progress tracking** using regular video uploads, which is essential for continuous emotional support.

Table 1: Comparison of Previous Researches

Criteria	Previous Research1(Ekman & Friesen, 1978) [1]	Previous Research 2(Zhu et al., 2017) [2]	Previous Research3(Schuller et al., 2011) [3]	Your Project
Facial Expression Recognition	Yes	Yes	No	Yes
Voice Tone Analysis for Emotion Detection	No	Yes	Yes	Yes
Multimodal Emotion Detection (Facial + Voice)	No	No	No	Yes
Real-time Emotion Detection	Yes	No	Yes	Yes
Personalized Emotional Regulation	No	Yes	Yes	Yes
Cultural Sensitivity Consideration	No	No	Yes	Yes
Ethical Handling of Emotional Data	No	Yes	No	Yes
Mobile App Integration for Emotional Tracking	No	No	Yes	Yes
Long-term Emotional Progress Tracking	No	No	No	Yes
Real-time Suggestions/Interventions	No	Yes	Yes	Yes

Table 2: Problems and Outputs

Problem Addressed	Technique	Analytical Output Parameters
Inaccurate Emotion Detection Using Single Modality	Multimodal Emotion Detection (Facial + Voice)	Accuracy of emotion detection (Happiness, Sadness, etc.)
Real-time Emotion Detection	Deep Learning Models (CNNs for Facial, LSTMs for Voice)	Real-time emotion classification and processing time
Personalized Emotional Regulation	AI-based Personalized Suggestions	Tailored interventions (coping strategies, resources)
Lack of Long-term Monitoring and Tracking	Mobile App Integration for Daily Facial Video Uploads	Emotional progress over time, consistency of emotion states
Cultural Insensitivity in Emotion Recognition	Culturally Sensitive Emotion Recognition Models	Cross-cultural accuracy, emotion recognition variance
Privacy and Ethical Concerns in Emotion Data Handling	Ethical Data Handling, Consent-based Framework	Data security, user consent, and privacy compliance
Lack of Real-time Feedback for Emotional Support	Real-time Feedback Mechanism (via App)	Frequency of feedback, user engagement

1.3 Research Problem

The primary research problem in this project is the development of an integrated AI system capable of accurately detecting and managing emotional states through a combination of facial expression recognition and voice tone analysis. Current emotion detection systems predominantly focus on either visual or auditory cues, which limits their ability to provide comprehensive and real-time emotional assessments. This project aims to combine both modalities for more accurate and reliable emotion detection. Additionally, there is a need for personalized, dynamic interventions based on detected emotions, as most existing systems lack real-time, adaptive feedback. Furthermore, challenges related to **cultural sensitivity** in emotion recognition, the **ethical handling of sensitive emotional data**, and **data privacy concerns** are not adequately addressed in many existing systems. The project will also explore long-term emotional progress tracking through regular facial video uploads, which is critical for continuous emotional support. This presents an opportunity to create a system that not only detects emotions but also offers ongoing, personalized emotional regulation interventions over time while ensuring privacy, ethical considerations, and inclusivity.

Table 3 : Research Descriptions

Research Element	Description
Research Problem	Developing an integrated AI system that combines facial expression recognition and voice tone analysis for accurate, real-time emotion detection and personalized interventions.
Focus of Existing Research	Existing research primarily focuses on either facial expression recognition or voice tone analysis for emotion detection, with limited exploration of combining both modalities for more accurate detection.
Identified Gap	Lack of multimodal emotion detection systems that integrate facial and voice cues in real-time. There is also a gap in providing personalized emotional regulation and long-term tracking.
Extended Research Problem	Expanding emotion detection systems by integrating both facial expression and voice tone analysis to create a real-time, adaptive system capable of offering personalized emotional interventions. Additionally, incorporating cultural sensitivity, ethical data handling, and long-term emotional tracking through mobile apps.
Key Ethical Concerns	Privacy concerns regarding the collection and use of sensitive emotional data, ensuring informed consent, and addressing the ethical implications of AI in mental health.
Proposed Solution	A system that combines facial expression and voice tone analysis to provide real-time emotion detection and personalized suggestions for emotional regulation, with mobile app integration for long-term emotional support. Ethical handling of data will be ensured through transparent consent processes and robust data security measures.
Objective Contribution	The project aims to contribute a comprehensive, AI-driven solution for emotion detection and regulation, enhancing emotional well-being through real-time feedback, personalized interventions, and long-term tracking. It will also address cultural, ethical, and privacy concerns, making the system more inclusive and responsible.

2 Objectives of the Project

2.1 Main Objective

The main objective of this project is to develop a comprehensive AI-driven system consisting of three key components: **hydration management**, **workout optimization**, and **mental health assessment**. The **hydration management** component aims to provide users with personalized fluid intake recommendations based on their body needs and activity levels. The **workout optimization** component focuses on designing adaptive exercise plans to enhance users' fitness levels while ensuring safety and efficiency. The **mental health assessment** component, which is the focus of this project, involves detecting and regulating emotional states through advanced AI techniques, offering support for mental well-being.

For the **mental health assessment** component, the goal is to create an AI system that integrates facial expression recognition and voice tone analysis to accurately detect the emotional states of users. By analyzing these visual and auditory cues, the system can predict emotions such as happiness, sadness, stress, and anger. Upon detecting an emotional state, the system will provide personalized suggestions, such as coping strategies, video resources, and other tools tailored to the user's needs. Additionally, the system will include a mobile application that allows for continuous emotional monitoring, tracking the user's progress over time and providing ongoing emotional support. Ethical handling of emotional data will be a priority, ensuring privacy, cultural sensitivity, and responsible use of data in supporting users' mental health.

2.2 Specific Objectives

2.2.1 Data Collection and Storage

To collect and store user-uploaded videos and voice recordings in a secure and ethical manner, ensuring user consent is obtained for emotional data processing and tracking. The data will be stored in a secure database, maintaining privacy and confidentiality.

2.2.2 Data Preprocessing and Feature Engineering

To preprocess the collected data by extracting relevant features from facial expressions (using CNNs) and voice tone (using LSTMs), transforming the data into a format suitable for machine learning model training. This step will involve noise reduction, normalization, and feature selection to improve model performance.

2.2.3 Machine Learning Model Development

To develop and train deep learning models for emotion detection, combining facial expression recognition and voice tone analysis. The models will be fine-tuned using data augmentation, transfer learning, and other techniques to enhance accuracy and generalization.

2.2.4 Demographic Categorization

To categorize users based on demographic information (such as age, gender, and cultural background) to improve emotion detection accuracy and ensure personalized, culturally sensitive

suggestions. This will allow the system to adapt and provide more relevant emotional support for diverse user groups.

2.2.5 Personalized Workout Plan Generation

To design a personalized emotional regulation plan for each user, based on their detected emotional state. This may include coping strategies, self-help exercises, and other resources to promote emotional well-being. The plan will evolve over time based on user feedback and emotional progress.

2.2.6 Testing and Validation

To conduct rigorous testing of the emotion detection system using real-world data, ensuring that it provides accurate emotion recognition and personalized interventions. This phase will also include validation of the mobile app integration and its functionality in providing continuous emotional support.

2.2.7 Ethical Considerations

To ensure that the project adheres to ethical standards, particularly regarding user consent, data privacy, and the responsible use of sensitive emotional data. The system will be designed with transparency and user control over the data being collected.

2.2.8 Integration and Deployment

To integrate the emotion detection system with a mobile application that allows for regular video uploads, continuous emotional tracking, and personalized feedback. The system will be deployed on both web and mobile platforms to ensure accessibility and ease of use for all users.

2.2.9 User Feedback and Iterative Refinement

To collect user feedback on the system's performance and emotional support effectiveness, and refine the system iteratively based on this feedback. This will ensure that the system continues to meet the users' needs and improves over time.

2.2.10 Evaluation of Ethical Impact

To evaluate the ethical impact of the emotion detection and regulation system, particularly focusing on privacy concerns, the potential for misuse of emotional data, and ensuring fairness and inclusivity in the system's recommendations and interventions.

3. Methodology

3.1 Proposed Project Execution

The proposed project execution involves several stages, starting with the collection and preprocessing of data, followed by the development of machine learning models for emotion detection. The system will be designed to analyze user-uploaded videos to detect emotional states based on facial expressions and voice tone. The project will then focus on providing personalized emotional regulation interventions through the mobile app, which will track the user's emotional progress over time. Ethical guidelines regarding data privacy and cultural sensitivity will be incorporated at every stage of the project to ensure responsible data handling. The final stage will

include extensive testing and validation of the system's functionality, accuracy, and ethical compliance.

3.2 Tasks and Sub-tasks

1. **Data Collection and Preparation**
    - Sub-task 1: Collect user-uploaded facial videos and voice recordings.
    - Sub-task 2: Obtain user consent and ensure privacy.
    - Sub-task 3: Preprocess videos (e.g., facial landmark detection, noise removal).
2. **Emotion Detection Model Development**
    - Sub-task 1: Develop and train CNNs for facial expression recognition.
    - Sub-task 2: Develop and train LSTMs for voice tone analysis.
    - Sub-task 3: Integrate both models into a multimodal emotion detection system.
3. **Personalized Emotional Regulation**
    - Sub-task 1: Design personalized coping strategies and emotional feedback.
    - Sub-task 2: Implement real-time intervention suggestions based on detected emotional states.
4. **Mobile Application Integration**
    - Sub-task 1: Develop the mobile app for daily video uploads and emotional progress tracking.
    - Sub-task 2: Integrate feedback mechanisms and reminders for emotional regulation.
5. **Testing and Validation**
    - Sub-task 1: Test system performance and accuracy using real-world data.
    - Sub-task 2: Conduct user feedback collection and refinement processes.
6. **Ethical Considerations and Privacy**
    - Sub-task 1: Implement data security measures.
    - Sub-task 2: Ensure transparent user consent processes.

3.3 Required Materials

- **Hardware:**
    - High-performance computers or servers for training deep learning models.
    - Cameras (smartphones, webcams) for video collection or sample data.
- **Software:**

    - Python programming language with libraries like TensorFlow, Keras, OpenCV for facial recognition, and PyTorch for LSTM implementation.
    - Mobile app development frameworks (e.g., Flutter, React Native) for app deployment.
- **Datasets:**
    - Emotion-labeled facial expression datasets (e.g., FER2013, AffectNet).
    - Voice emotion datasets for tone analysis (e.g., RAVDESS).
- **Tools:**
    - Data processing tools (e.g., Pandas, NumPy).
    - Cloud storage solutions for data storage and sharing.

3.4 Time Frame

Stage	Time Frame
Data Collection and Preprocessing	1 month
Model Development (CNNs & LSTMs)	2 months
Personalized Emotional Regulation Design	1 month
Mobile Application Development	2 months
Testing and Validation	1 month
Ethical Considerations and Privacy Setup	Ongoing throughout project
Final Testing, Deployment, and Refinement	1 month

3.5 Schedules

- **Task Charts:**
  Gantt charts will be used to visualize the timeline and dependencies of key tasks, ensuring a structured workflow. The chart will track progress across phases such as data collection, model development, app integration, and testing.
- **Team Collaboration:** The project team is divided into specialized roles:
- **Frontend/UI Developer:** Designs the mobile app interface.
- **Backend Developer:** Manages server-side logic and app integration.
- **Machine Learning Engineer:** Develops and trains the emotion detection models.
- **Database Manager:** Manages secure data storage and retrieval.

3.6 Anticipated Conclusion

The anticipated conclusion of this project is the successful development of a comprehensive emotion detection and regulation system that leverages both facial expression recognition and voice tone analysis to offer real-time emotional support. The mobile app will provide continuous tracking of emotional progress, delivering personalized interventions and coping strategies based on the user's emotional state. The system will also ensure that ethical considerations regarding user privacy, consent, and data security are met throughout the project's development. Upon completion, the system will provide a practical solution for emotional well-being management and contribute to advancing AI-based mental health support tools.

4. Project Requirements

4.1 Functional Requirements

- **Emotion Detection:**The system should be able to accurately detect emotional states (e.g., happiness, sadness, anger, stress) from user-uploaded videos by analyzing facial expressions and voice tones.

- **Real-Time Processing:** The emotion detection system should provide real-time results as soon as the video is uploaded.

- **Personalized Interventions:** Based on the detected emotional state, the system should provide personalized suggestions (e.g., coping strategies, video links) to help the user manage their emotions.

- **Mobile App Integration:** The system must integrate with a mobile application that allows users to upload daily facial videos for continuous emotional tracking.

- **Progress Tracking:** The mobile app should track the user’s emotional progress over time and offer real-time feedback and reminders for emotional regulation.

- **Multimodal Emotion Detection:** The system must integrate both facial expression recognition (using CNNs) and voice tone analysis (using LSTMs) to provide a comprehensive emotional assessment.

- **User Feedback System:** The system should allow users to provide feedback on the accuracy and usefulness of the emotional support offered.

- **Data Security:** Ensure secure data storage and processing of user-uploaded videos and emotional data, including proper encryption methods and user consent for data use.

4.2 Non-Functional Requirements

- **Performance:** The emotion detection system must provide results within 10 seconds after video upload, ensuring a responsive user experience.
- **Scalability:** The system should be able to scale to support multiple users simultaneously without a decrease in performance.

- **Usability:** The mobile app should have an intuitive user interface (UI) to make it easy for users to upload videos, track emotional progress, and access interventions.

- **Reliability:** The system must be stable, with minimal downtime, and ensure that video uploads, emotion detection, and feedback are consistently operational.

- **Security:** The system must follow industry standards for data security, ensuring that all user data is stored and transmitted securely, and user privacy is respected.

- **Compliance:** The system should comply with relevant data privacy laws, such as GDPR, for handling user data.

4.3 System Requirements

The system requires high-performance hardware, such as a server or cloud platform, for training and processing deep learning models. Devices like smartphones and webcams will capture user videos. Python will be used for deep learning, with libraries like TensorFlow, Keras, and PyTorch. The system will use MySQL or MongoDB for secure data storage and React Native or Flutter for mobile app development. A stable internet connection is necessary for real-time video uploading and processing, while cloud storage will securely store both video and emotional data, ensuring scalability and data security across platforms.

4.4 User Requirements

The system must allow users to register securely using an account with a username, password, and email, and must provide informed consent for data collection. Users should be able to easily upload videos, whether facial or full-body, for emotion detection. Once the emotional state is detected, the system should provide personalized emotional regulation suggestions, including coping strategies or video links tailored to the user's needs. Additionally, users should be able to monitor their emotional progress over time through the mobile app, receiving reminders and feedback to support their emotional well-being. Privacy control is a critical feature, enabling users to manage what data is shared, access their data, and request its deletion. The system

should also ensure accessibility for users from diverse cultural backgrounds, ensuring that emotional feedback is relevant, inclusive, and culturally sensitive.

4.5 System Diagram

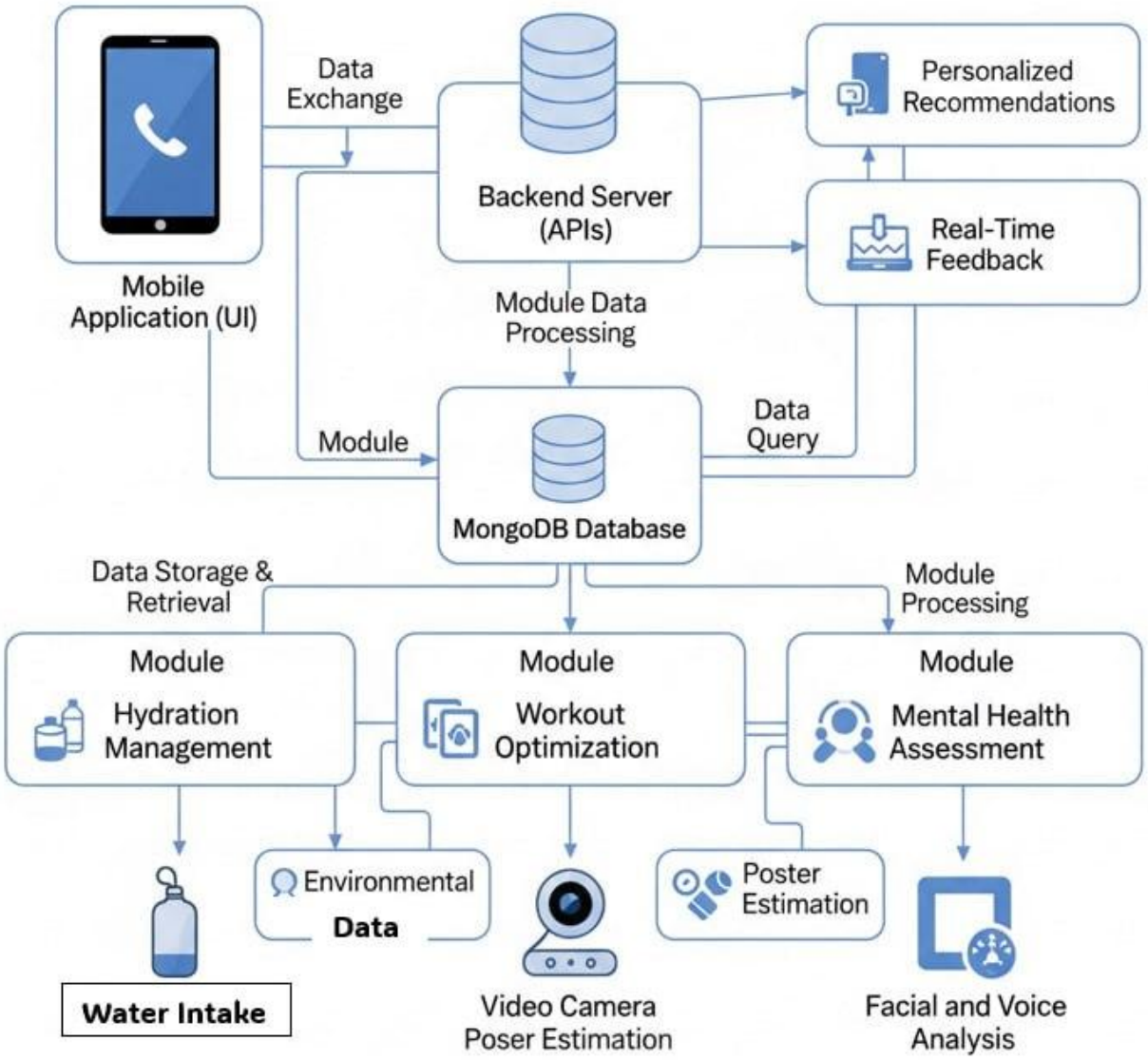
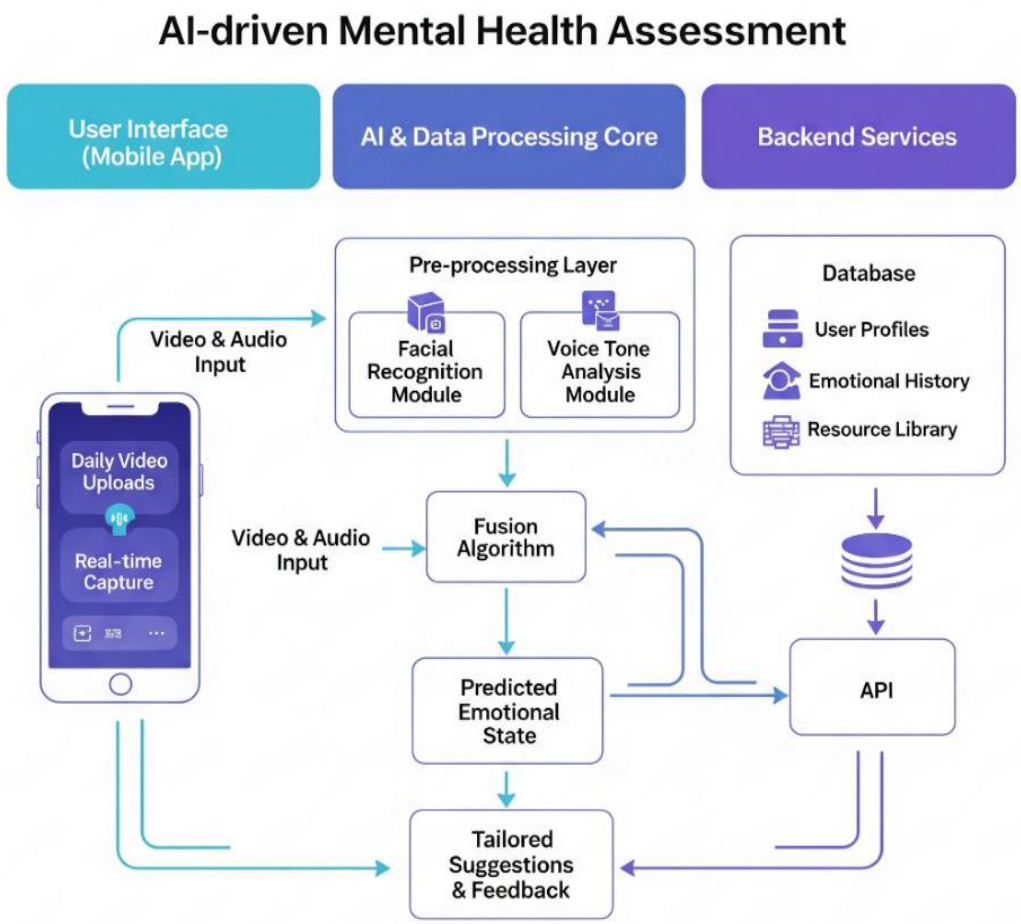


Figure10: System Diagram

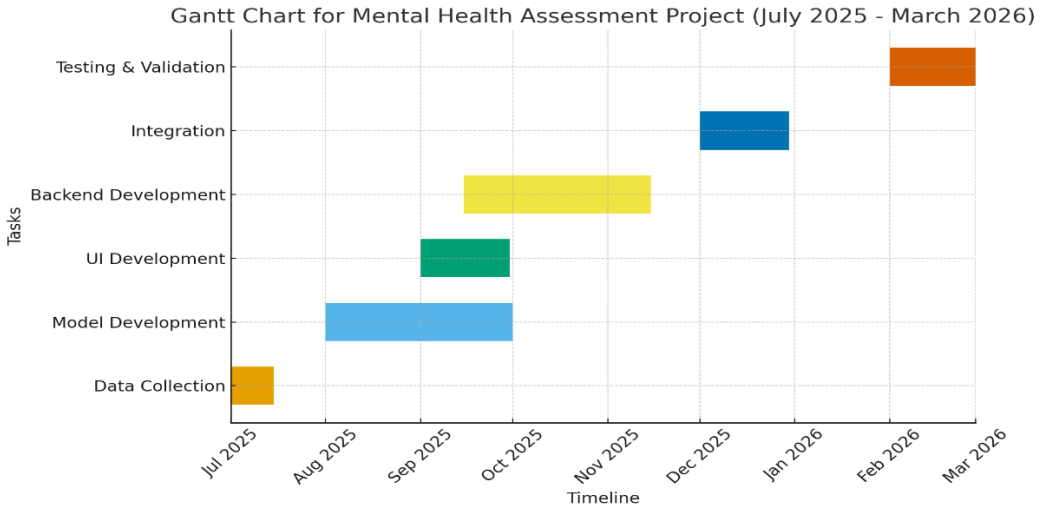
4.6 Component Diagram

Figure11: Component Diagram



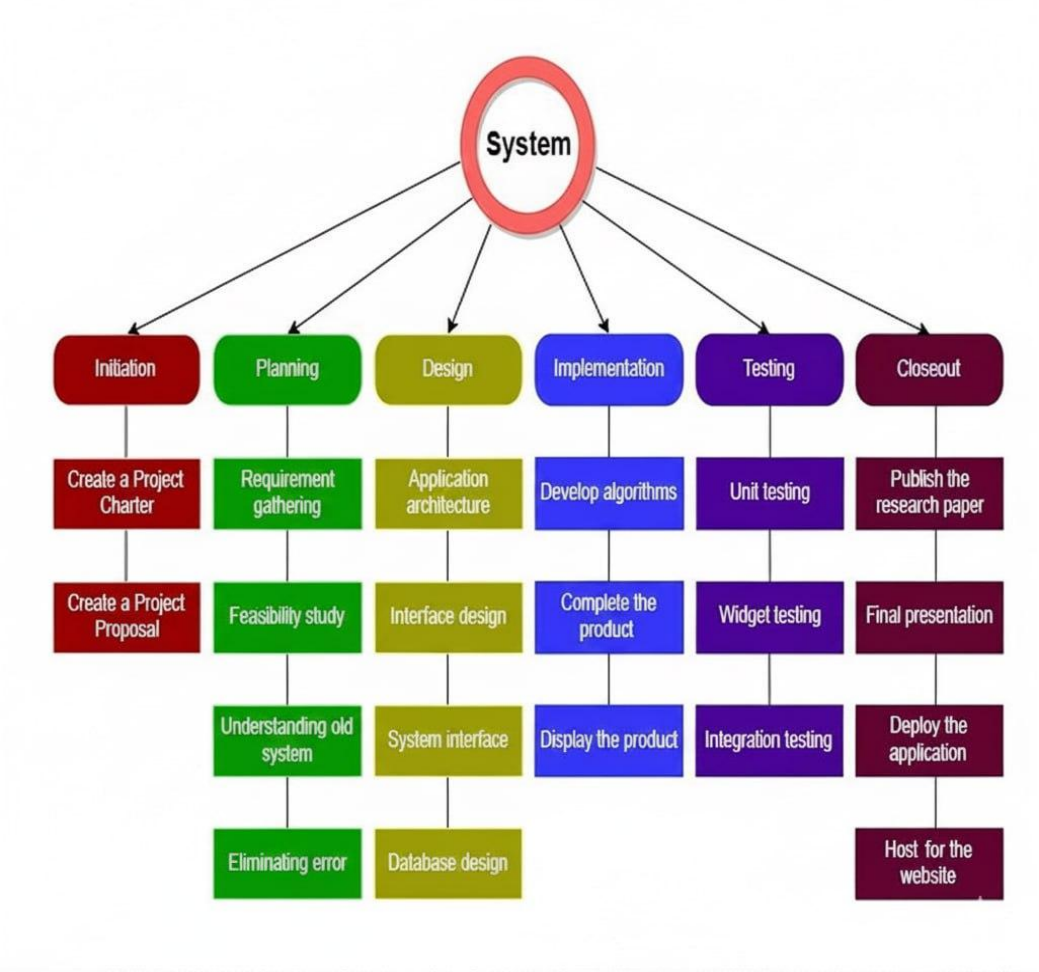
4.7 Gant Chart

Figure12: Gant Chart



4.8 Work Breakdown Chart

Figure13: Work Breakdown Chart



5.BUDGET AND BUDGET JUSTIFICATION

5.1 Budget

- ✓ Cloud Services LKR 40,000
- ✓ MongoDB Atlas LKR 15,000 annual
- ✓ Advertisement LKR 1,500
- ✓ Development Tools LKR 2,000

5.2 Budget Justification

1.Cloud Services (LKR 40,000 annually):

The largest portion of the budget is allocated for cloud services like AWS, GCP, or Azure. These services provide essential resources such as compute power (EC2, Lambda), data storage, and other necessary cloud infrastructure to support the AI-driven hydration management system. The cloud services are crucial for scaling the system as user demand increases and for ensuring high availability and performance

2.MongoDB Atlas (LKR 15,000 annually):

MongoDB Atlas is selected for its efficiency in managing large volumes of unstructured data, such as user profiles, hydration history, and environmental data. The annual cost includes database hosting, backups, and security services, ensuring that the database is accessible, secure, and scalable.

3. Advertisement (LKR 1,500/month):

A portion of the budget is dedicated to social media advertising on platforms like Facebook and Instagram. This helps increase awareness of the platform after launch and allows for targeted audience to reach maximize user acquisition. These platforms offer affordable paid post boosting options that will drive traffic to the mobile application or website.

4.Development Tools (LKR 2,000 annual):

Minimal expenses are allocated for development tools and software libraries essential for coding and machine learning model development. This includes IDEs, version control services (like GitHub), and other utilities to support the backend and mobile app development.

5.3 Commercialization

1. Subscription Model

- ✓ Overview: A subscription-based pricing model will be implemented to offer users various pricing tiers.
- ✓ Basic Plan: Provides essential features, such as basic hydration tracking and general recommendations.
- ✓ Premium Plan: Offers advanced features, such as personalized hydration plans, daily recommendations based on detailed activity logs, and in-depth health analytics.

2. Revenue Generation: Monthly or yearly subscriptions will generate recurring revenue, ensuring consistent cash flow and incentivizing users to stay engaged with the platform over time

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7.Appendices

Appendix A - Survey Link

https://docs.google.com/forms/d/e/1FAIpQLSesB-SK3TsE2gUQA5GF-BjH7OQvyVx5oP_ykfZBaMqQAwLIyA/viewform?usp=dialog